

Ch. 14: Ideal & Quotient Rings

Def. A subring I in a ring R is a (two-sided) ideal of R if for every $r \in R$ & every $a \in I$ the products ra & ar are in I .

Multiplicative
Absorption
Property

The whole ring is always an ideal of itself, so we say an ideal I of R is proper if $I \subsetneq R$.

Examples: $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

$\{\bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{0}\}$ is an ideal of $\mathbb{Z}/10\mathbb{Z}$.

$$\mathbb{Z}[i] = \{a+bi : a, b \in \mathbb{Z} \text{ \& } i^2 = -1\}$$

let $I = \{\alpha \in \mathbb{Z}[i] : \alpha = \beta \cdot (1-i) \text{ for some } \beta \in \mathbb{Z}[i]\}$.

It turns out I is an ideal.

Important Example: let R be a commutative r.-ing.

The set $\langle a \rangle \stackrel{\text{def}}{=} \{ra : r \in R\} = \{\text{all the multiples of } a\}$ is an ideal of R . It is called the principal ideal generated by

Thm 14.1 (The Ideal Test) $\xrightarrow{\text{generated by one element}}$ $\underline{a}$.

A nonempty subset I of a ring R is an ideal if

1. $a-b \in I$ $\forall a, b \in I$ (I is a subgroup of $(R, +)$)

2. ra & ar are in I $\forall r \in R$ and $a \in I$.

(Mult. absorption property stronger than closure under mult.)
suppose $I \subset R$ is a non-empty subset that satisfies properties 1. & 2.

Proof. We see $a-b \in I$ and $ab \in I$ $\forall a, b \in I$.

Thus I is a subring of R . Since 2. is exactly the mult. absorption property and since I is a subring of R , we see I is an ideal of R .

Ex. $\langle n \rangle$

$n\mathbb{Z}$ is an ideal of $\mathbb{Z}$. Clearly, $n\mathbb{Z}$ is nonempty

If $a, b \in n\mathbb{Z}$, we see $a = nk$ & $b = nm$ for some $k, m \in \mathbb{Z}$.

Thus $a-b = nk - nm = n(k-m) \in n\mathbb{Z}$.

If $r \in \mathbb{Z}$, then $ra = ar = nkr \in n\mathbb{Z}$.

Thus the ideal test shows $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$!

Claim. $\langle a \rangle$ is an ideal of R .

Proof. Let $x, y \in \langle a \rangle$. We see $x = a \cdot s$ & $y = a \cdot t$ for some $s, t \in R$. Thus $x - y = as - at = a(s - t) \in \langle a \rangle$

Let $r \in R$. We see $ra = ar \in \langle a \rangle = \{ar : r \in R\}$.

Thus the ideal test shows $\langle a \rangle$ is an ideal of R .

Quotient Rings. Let R be a ring

and I an ideal of R . Write

$R/I = \{r + I : r \in R\}$ for the set

of cosets of I in R . Define

addition $(r + I) + (s + I) = (r + s) + I \quad \forall r, s \in R$.

We already know that $(R/I, +)$ is an Abelian Group

Define multiplication $(r + I)(s + I) = rs + I \quad \forall r, s \in R$.

Theorem 14.2. R/I is a ring. In fact, if A is a subring of R

R/A is a ring $\iff A$ is an ideal.

$f, \cdot$ are well-defined. $(R/I, +)$ is an Abelian group, $\cdot$ is associative.
 Closure under $f, \cdot$. Left & Right Distributivity

Proof. From our work with groups we know $(R/I, +)$ is an Abelian group. $\checkmark$

If $r+I, s+I, t+I$ are in R/I , we check

$$\begin{aligned} (r+I)(s+I)(t+I) &= (rs+I)(t+I) = (rs)t+I \\ &\stackrel{\substack{\uparrow \\ \cdot \text{ is associative in } R}}{=} r(st)+I = (r+I)(st+I) = (r+I)((s+I)(t+I)) \end{aligned}$$

yay!

Left & Right Distributivity are also inherited from R .

Interesting part: Mult. is well-defined.

Suppose $r+I = r'+I$ & $s+I = s'+I$

We want $(r+I)(s+I) = rs+I$ is the same as
 as $(r'+I)(s'+I) = r's'+I$.

Since $r+I = r'+I$ & $s+I = s'+I$,

We have $r' = r+a$ & $s' = s+b$ for some $a, b \in I$.

Ex. $1+4\mathbb{Z} = 81+4\mathbb{Z}$ $1 = 81-80$ & $80 \in 4\mathbb{Z}$

$$r's' = (r+a)(s+b) = rs + \underbrace{as + rb + ab}_{\in I}$$

Since I is an ideal and $a, b \in I$,

$as \in I$ and $rb \in I$ (mult. absorption)

Thus $as + rb + ab \in I$.

Hence $r's' + I = rs + I$. ✓

Prime Ideals $\hat{=}$ Maximal Ideals

(positive integers)

Euclid's lemma. p is prime if and only if whenever p divides a product ab , $p|a$ or $p|b$.

A proper ideal P in a commutative ring R is called prime if whenever $ab \in P$, then $a \in P$ or $b \in P$. "To contain is to divide"

A maximal ideal M of a commutative ring R is a proper ideal of R such that whenever M is contained in an ideal I of R , then $I = M$ or $I = R$.

$[M \subsetneq I \subsetneq R \text{ is impossible}]$

Thm. 14.3. Let R be a comm. ring with 1, and let A be an ideal of R .

R/A is an integral domain $\Leftrightarrow A$ is prime.

Thm. 14.4. Let R be a comm. ring with 1, and let A be an ideal of R .

R/A is a field $\Leftrightarrow A$ is maximal.

Example. Prove $\langle x \rangle \subset \mathbb{Z}[x]$ is a prime ideal that is not maximal.

Consider $\mathbb{Z}[x]/\langle x \rangle$. "When quotienting everything in the ideal becomes 0"

$$\mathbb{Z}[x]/\langle x \rangle = \mathbb{Z}[0] \cong \mathbb{Z}$$

Let $a_n x^n + \dots + a_1 x + a_0 + \langle x \rangle$ be an arbitrary coset.

We see $a_n x^n + \dots + a_1 x \in \langle x \rangle$, so

$$a_n x^n + \dots + a_1 x + a_0 + \langle x \rangle = a_0 + \langle x \rangle. \text{ Thus}$$

every coset has the form $a + \langle x \rangle$ for $a \in \mathbb{Z}$.

If $a \neq b$, then $a + \langle x \rangle \neq b + \langle x \rangle$ since

$a - b \notin \langle x \rangle$. Thus $\mathbb{Z}[x]/\langle x \rangle$ is pretty much the same as $\mathbb{Z}$.
isomorphic

Since $\mathbb{Z}$ is an integral domain,

we see $\langle x \rangle$ is prime (Thm. 14.3)

Since $\mathbb{Z}$ is not a field, we see $\langle x \rangle$ is not maximal (Thm. 14.4)

Thm. 14.3. Let R be a comm. ring with 1,
and let A be an ideal of R .

R/A is an integral domain $\Leftrightarrow A$ is prime.

Proof. ($\Leftarrow$) Suppose A is prime. Let $x+A, y+A$ be
arbitrary elements of R/A . (In other words, $x, y \in R$ are arbitrary)

Suppose $(x+A)(y+A) = 0+A$. By def. $(x+A)(y+A) = xy+A$.

Thus $xy+A = 0+A$. This implies $xy \in A$. Since A
is prime $x \in A$ or $y \in A$. Thus $x+A = 0+A$ or $y+A = 0+A$. $\checkmark$

($\Rightarrow$) Suppose R/A is an integral domain. Let
 $x, y \in R$ such that $xy \in A$. Thus $xy+A = 0+A$.
Hence $(x+A)(y+A) = 0+A$. Since R/A is an integral
domain, $x+A = 0+A$ or $y+A = 0+A$. Thus $x \in A$ or $y \in A$. $\square$

Thm. 14.4. Let R be a comm. ring with 1,
and let A be an ideal of R .

R/A is a field $\Leftrightarrow A$ is maximal.

Proof. ($\Rightarrow$) Suppose R/A is a field and suppose B is an ideal of R such that $A \subsetneq B$. Let $b \in B - A = B \setminus A$. Thus $b + A$ is a non-identity element of R/A . Since R/A is a field $\exists c + A \in R/A$ such that $(b + A)(c + A) = 1 + A$. Thus $bc + A = 1 + A$.

Hence $bc + a = 1$ for some $a \in A$. Now B is an ideal containing A . Thus $a \in B$ and $bc \in B$, so $bc + a \in B$. I.e., $1 \in B$. This implies $B = R$. Indeed, if $r \in R$, then $r \cdot 1 = r \in B$, so $R \subseteq B$ and $R = B$. We see that A is maximal.

($\Leftarrow$) Suppose A is maximal. Let $b \in R - A$. We wish to show $b + A$ has a multiplicative inverse in R/A .

Consider the set $B = \{br + a : r \in R, a \in A\}$.

We claim* this is an ideal of R that properly contains A . Since A is maximal, we see $B = R$. In particular, $1 \in B$. Thus $1 = br + a$ for some $r \in R$ and some $a \in A$.

Thus $1 + A = br + A = (b + A)(r + A)$. Hence $b + A$

has a multiplicative inverse in R/A . $\square$

Ch. 15: Ring Homomorphisms and the First Isomorphism Theorem

A ring homomorphism is a function $\phi: R \rightarrow S$ between rings $R \ni S$ such that

$$\phi(a+b) = \phi(a) + \phi(b) \quad \& \quad \phi(ab) = \phi(a) \cdot \phi(b) \quad \forall a, b \in R.$$

" ϕ respects the ring operations."

If in addition ϕ is bijective, then we say ϕ is a ring isomorphism. In this case the structure of S is the same as that of R and we write $R \cong S$.

Ex: $\mathbb{Z}[i]/\langle i-1 \rangle \cong \mathbb{Z}/2\mathbb{Z}$ from homework.

$$\mathbb{R}[x]/\langle x^2+1 \rangle \cong \mathbb{C}$$

Examples of homomorphisms:

$$\text{red. } \mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$$

$$a \mapsto \overline{a} = a + m\mathbb{Z} = a + \langle m \rangle$$

$$\iota: \mathbb{Z} \hookrightarrow \mathbb{Q}$$

$$a \hookrightarrow a$$

$$[a, 1]$$

$$\text{Eval}_1: \mathbb{Z}[x] \rightarrow \mathbb{Z}$$

$$f(x) \mapsto f(1)$$

$$x \mapsto 1$$

$$\text{Ker Eval}_1$$

$$\{f(x) \in \mathbb{Z}[x] : f(1) = 0\}$$

$$x^3 - 1 \mapsto 1^3 - 1 = 0 \quad \{f(x) \in \mathbb{Z}[x] : (x-1) \mid f(x)\}$$

If $\phi: R \rightarrow S$ is a ring hom. define the kernel of ϕ

$$\text{ker } \phi = \{r \in R : \phi(r) = 0_S\}$$

The image of a hom. $\phi: R \rightarrow S$ is

$$\phi(R) = \{s \in S : \phi(r) = s \text{ for some } r \in R\}.$$

Theorems 15.2 & 15.4: $\ker \phi$ is an ideal and every ideal looks like $\ker \phi$ for some hom. ϕ .

Thm. 15.3. (The first isomorphism theorem for Rings)

If $\phi: R \rightarrow S$ is a ring hom., then

$$R/\ker \phi \cong \phi(S)$$

Ex. Define $\phi: \mathbb{Z}[x] \rightarrow \mathbb{Z}[i]$ by $\phi(x) = i$. We can see ϕ is surjective.

$$(a+bx \mapsto a+bi)$$

$$\ker \phi = \{f(x) \in \mathbb{Z}[x] : f(i) = 0\}$$

With some work we can show $\ker \phi = \langle x^2 + 1 \rangle$.

$$\text{F.I.T.} \quad \mathbb{Z}[x] / \langle x^2 + 1 \rangle \cong \mathbb{Z}[i]$$

Thm. 15.1.4. If $\phi: R \rightarrow S$ is a ring hom. and $B \subset S$ is an ideal, then

$\phi^{-1}(B) = \{r \in R: \phi(r) \in B\}$ is an ideal of R

Proof. Since B is an ideal, $0_S \in B$. Since

$\phi: (R, +) \rightarrow (S, +)$ is a group hom. $\phi(0_R) = 0_S$.

Thus $0_R \in \phi^{-1}(B)$, so $\phi^{-1}(B) \neq \emptyset$. $\checkmark$

Let $x, y \in \phi^{-1}(B)$, and consider $\phi(x - y) \stackrel{\phi(x + (-y)) = \phi(x) + \phi(-y) = \phi(x) - \phi(y)}{=} \phi(x) - \phi(y)$.

Since $\phi(x), \phi(y) \in B$ and B is an ideal, $\phi(x) - \phi(y) \in B$.

Thus $\phi(x - y) \in B$. $\checkmark$

Let $r \in R$ and $x \in \phi^{-1}(B)$. We consider rx . We

want $rx \in \phi^{-1}(B)$. Thus we want $\phi(rx) \in B$.

Now since ϕ respects mult. $\phi(rx) = \phi(r) \cdot \phi(x)$.

Since $\phi(x) \in B$ and B has mult. absorption, $\phi(r)\phi(x) \in B$.

Thus $rx \in \phi^{-1}(B)$. $\square$

Cor.^{15.2} $\ker \phi$ is an ideal.

Proof. Note $\ker \phi = \{r \in R: \phi(r) = 0_S\} = \phi^{-1}(0)$

and $\{0_S\}$ is an ideal of S .

Thm. 15.1.2. $\phi(R)$ is a subring of S .

"
 $\{s \in S: s = \phi(r) \text{ for some } r \in R\}$

Student
Debt.

Proof. $0_R \in R$ and $\phi(0_R) = 0_S$, so $\phi(R) \neq \emptyset$.
Let $u, v \in \phi(R)$ and consider $u - v$.

Prove that
if F is
a field &

Now $u = \phi(a)$ & $v = \phi(b)$ for some $a, b \in R$.

$I \subset F$ is
an ideal, then
 $I \in \{0, F\}$.

Thus $u - v = \phi(a) - \phi(b) = \phi(a - b)$.

Since $a - b \in R$, $\phi(a - b) = u - v \in \phi(R)$. ☺

Consider $u \cdot v = \phi(a) \phi(b) = \phi(ab)$, since ϕ is a ring hom.

Thus $uv \in \phi(R)$, and we conclude $\phi(R)$ is a subring of S . $\square$

Ex.

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\text{injective}} & \mathbb{Q} \\ a & \mapsto & a \end{array}$$

Thus $\mathbb{Z} = \mathcal{L}(\mathbb{Z})$
is a subring of $\mathbb{Q}$.

Note it is not an ideal.

F.I.T. If $\phi: R \rightarrow S$ is a ring hom., then

$R/\ker\phi \cong \phi(R)$ with the isomorphism
given by $r + \ker\phi \xrightarrow{\bar{\phi}} \phi(r)$.

Proof. We have already seen that $R/\ker\phi \cong \phi(R)$
as additive groups. Thus we just need to show that
 $\bar{\phi}$ respects mult. let $r + \ker\phi, s + \ker\phi \in R/\ker\phi$.

Then $\bar{\phi}((r + \ker\phi)(s + \ker\phi)) = \bar{\phi}(rs + \ker\phi) = \phi(rs)$.

Now $\phi(rs) = \phi(r)\phi(s) = \bar{\phi}(r + \ker\phi) \cdot \bar{\phi}(s + \ker\phi)$.

Thus $\bar{\phi}((r + \ker\phi)(s + \ker\phi)) = \bar{\phi}(r + \ker\phi) \cdot \bar{\phi}(s + \ker\phi)$.

Example. $\mathbb{Z}[\varepsilon] = \{a + b\varepsilon : a, b \in \mathbb{Z}, \varepsilon^2 = 0\}$ ☺

My friend

$\mathbb{Z}[x]$. Define $\phi: \mathbb{Z}[x] \rightarrow \mathbb{Z}[\varepsilon]$

by $\phi(x) = \varepsilon$. I.e. $\phi(f(x)) = f(\varepsilon)$.

$\phi(\mathbb{Z}[x]) = \mathbb{Z}[\varepsilon]$, so ϕ is surjective.

$\mathbb{Z}[x] \rightarrow \mathbb{Z}[\varepsilon]$ one can show $\ker\phi = \langle x^2 \rangle$

$$\text{F.I.T. } \mathbb{Z}[x]/\langle x^2 \rangle \cong \mathbb{Z}[\varepsilon]$$

Since $x \cdot x \in \langle x^2 \rangle$ but $x \notin \langle x^2 \rangle$, we see $\langle x^2 \rangle$ is not prime.

Sure enough $\varepsilon \cdot \varepsilon = 0$, so we have zero divisors

in $\mathbb{Z}[x]/\langle x^2 \rangle$ we have $x \cdot x = 0$

$$\mathbb{Z}[x] \xrightarrow{\psi} \mathbb{Z}[i] \quad x \mapsto i$$

$$f(x) \mapsto f(i)$$

$$\psi(\mathbb{Z}[x]) = \mathbb{Z}[i], \quad \ker \psi = \langle x^2 + 1 \rangle$$

$$\text{F.I.T. } \mathbb{Z}[x]/\langle x^2 + 1 \rangle \cong \mathbb{Z}[i]$$

$$x^3 - 3x^2 + 2x + 7 \in \langle x^2 + 1 \rangle \stackrel{?}{=} a + bi$$

*Think of stuff in the ideal as 0. $x^2 + 1 = 0$
 $x^2 = -1$

$$x(-1) - 3(-1) + 2x + 7 \in \langle x^2 + 1 \rangle$$

$$x + 10 \in \langle x^2 + 1 \rangle \xrightarrow{x \mapsto i} i + 10$$